PlanSmartAI

RAG Pipeline — R&D Weekly Report

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1. Project Overview

PlanSmartAI is a multimodal Retrieval-Augmented Generation (RAG) pipeline built specifically for construction documents — engineering blueprints and specification files. The system lets users upload PDFs and ask natural-language questions against them, getting structured answers with cited page sources and confidence scores.

This report documents three weeks of R&D work: framework selection (Week 1), baseline pipeline and HuggingFace model research (Week 2), and the full production-grade pipeline with visual tiling, hybrid search, and agentic orchestration (Week 3).

Project: **PlanSmartAI — Construction Document Intelligence**

Organization: **2build.ai**

Report Period: **Weeks 1–3, March 2026**

Pipeline Type: **Multimodal RAG (Visual + Text, Hybrid Search)**

Document Domain: **Engineering Blueprints & Construction Specifications**

2. Week 1 — Framework Selection & Architecture Decisions

2.1 Why LangGraph + LangChain

The first week was entirely about evaluating what orchestration framework to build the agentic pipeline on. After reviewing the landscape of agent frameworks — including bare LangChain chains, CrewAI, AutoGen, and LangGraph — LangGraph was selected as the foundation.

Key reasons for choosing LangGraph:

- Explicit stateful graph with typed state objects — critical for RAG pipelines where state (retrieved docs, confidence, retry count) must persist across nodes
- Native support for conditional routing (confidence_check → rewrite vs generate) without brittle if/else chains
- Parallel branch execution — blueprint_retrieve and text_retrieve can run concurrently, with operator.add reducer safely merging results
- First-class support for retry loops with cycle detection
- LangChain integration — reuses LangChain's LLM wrappers, text splitters, and document abstractions

LangChain was selected alongside LangGraph for its:

- ChatOpenAI wrapper — works with any OpenAI-compatible API including OpenRouter, enabling access to Claude 3.5 Sonnet
- RecursiveCharacterTextSplitter — section-aware chunking with heading-priority separators
- Document and message schema abstractions already expected by LangGraph nodes

2.2 Key Architecture Decisions Made in Week 1

Decision	Choice Made	Rationale
Agent framework	LangGraph 0.2.28	Stateful graph, parallel branches, retry loops
LLM abstraction	LangChain + langchain-openai	OpenAI-compatible, works with OpenRouter
Vision LLM	Claude 3.5 Sonnet via OpenRouter	Best multimodal accuracy for blueprint reading
REST API	FastAPI + Uvicorn	Async, fast, BackgroundTasks for ingestion
Chat UI	Chainlit 2.x	RAG-native UI with file upload + markdown rendering
Vector DB	Qdrant (Docker)	Supports MultiVector + Sparse — only DB that handles ColQwen2 natively

Decision	Choice Made	Rationale
Relational DB	PostgreSQL 15	Document metadata, page image paths, status tracking

3. Week 2 — Baseline Pipeline & HuggingFace Model Research

3.1 Baseline Text Retrieval Pipeline

Week 2 began by building a simple sample PDF analyzer to validate end-to-end text retrieval before adding any visual components. This established a working baseline: Docling parses PDFs to markdown, RecursiveCharacterTextSplitter chunks the text, and a basic dense vector search retrieves chunks.

The baseline immediately exposed the core problem: standard text chunking completely fails for engineering blueprints. A blueprint page converted to text becomes a stream of disconnected numbers, room labels, and annotations with no spatial context. The retriever could find individual dimension values, but had no understanding of which wall they described or where on the plan they appeared.

3.2 HuggingFace Model Research

The majority of Week 2 was spent evaluating HuggingFace models specifically suited for construction document analysis — prioritizing models that preserve spatial layout and geometric relationships rather than treating blueprints as unstructured text.

3.2.1 Visual Embedding: ColQwen2 Selected

ColQwen2 (vidore/colqwen2-v1.0) from the ColPali family was selected for blueprint page embedding. Unlike text-based embedders that discard visual layout entirely, ColQwen2 encodes each image as a set of patch-level vectors (multi-vector representation) rather than a single global embedding. At query time, Qdrant computes a MaxSim score — the maximum cosine similarity between each query patch vector and every document patch vector — preserving spatial correspondence between query concepts and their exact location on the blueprint.

This matters critically for blueprints: the question 'what is in the boardroom' needs to match the spatial region of the page where the boardroom appears, not just any page that mentions boardrooms.

3.2.2 Text Embedding: BGE-M3 Selected

BGE-M3 (BAAI/bge-m3) was selected for specification document embedding. Its key advantage is dual-representation output from a single model pass: a 1024-dimensional dense vector for semantic matching, and a sparse lexical weight dictionary for BM25-style exact-term matching. These two representations are fused at query time using Reciprocal Rank Fusion (RRF), which

consistently outperforms either representation alone on technical documents containing specific terminology, code references, and model numbers.

3.2.3 Reranker: BGE-Reranker-v2-M3 Selected

BGE-Reranker-v2-M3 (BAAI/bge-reranker-v2-m3) was selected as the cross-encoder reranker. After Qdrant retrieves a candidate set of pages or chunks, the reranker scores each query-document pair jointly (cross-encoder, not bi-encoder), producing much more precise relevance scores. Its output drives the confidence scoring that controls LangGraph's routing decisions — whether to proceed to generation, rewrite the query, or widen the search.

Model	HuggingFace ID	Purpose	Key Property
ColQwen2	vidore/colqwen2-v1.0	Blueprint visual embedding	Multi-vector patch-level, MAX_SIM late interaction
BGE-M3	BAAI/bge-m3	Spec text embedding	Dual dense+sparse from single model pass
BGE-Reranker-v2-M3	BAAI/bge-reranker-v2-m3	Cross-encoder reranking	Joint query-passage scoring for confidence
Claude 3.5 Sonnet	via OpenRouter	Vision LLM / answer generation	Multimodal, 200K context, instruction-following

3.2.4 Device Constraints on Apple Silicon

All three local models run on CPU. Apple Silicon MPS was tested for ColQwen2 but disabled — bfloat16 matmul shape incompatibilities in the ColQwen2 architecture cause runtime failures on MPS. Models run in float32 (ColQwen2) and fp16 (BGE-M3, Reranker) on CPU. This is the primary source of ingestion and query latency on the development machine.

4. Week 3 — Full Pipeline Build, Optimizations & Test Results

4.1 Complete System Architecture

The full production pipeline was built and tested end-to-end this week against a real 52-page construction blueprint PDF. The system handles two document types through separate pipelines that share a common query graph.

Layer	Technology	Version
Language	Python 3.11 (Homebrew)	—
REST API	FastAPI + Uvicorn	>=0.115.0 / >=0.30.0
Chat UI	Chainlit	>=2.0.0
Agent Orchestration	LangGraph	0.2.28
LLM Abstraction	LangChain + langchain-openai	0.2.16 / 0.1.23
LLM Gateway	OpenRouter (Claude 3.5 Sonnet)	—
Visual Embedding	ColQwen2 via colpali-engine	0.3.1
Text Embedding	BGE-M3 via FlagEmbedding	1.2.11
Reranker	BGE-Reranker-v2-M3 via FlagEmbedding	1.2.11
Vector DB	Qdrant (Docker)	1.17.0
Relational DB	PostgreSQL 15 (Docker)	15
ORM	SQLAlchemy + psycopg2-binary	2.0.35 / 2.9.9
PDF → Image	pdf2image + pypdfium2	1.17.0
PDF → Markdown	Docling	2.5.0
Image Processing	Pillow (PIL)	10.4.0
Deep Learning Runtime	PyTorch + torchvision	2.6.0 / 0.21.0

4.2 Blueprint Ingestion Pipeline

Stage 1 and Stage 2 run concurrently via ThreadPoolExecutor. Stage 1 converts every PDF page to JPEG using pdf2image at 250 DPI, quality 92, with pdftocairo as the backend. Stage 2 extracts raw text per page via pypdfium2. Both run in parallel under asyncio.gather.

Stage 3 runs ColQwen2 embedding in a dedicated CPU executor, batch size 2 for CPU stability. Each page produces a multi-vector embedding — one 128-dimensional vector per image patch.

Stage 4 upserts into Qdrant's blueprint_visual collection in batches of 5. Stage 5 bulk-inserts all page records into PostgreSQL.

4.3 Specification Document Ingestion Pipeline

Docling converts the PDF to structured markdown with table detection enabled and OCR disabled. RecursiveCharacterTextSplitter chunks the markdown at 1500 characters with 200-character overlap, using heading markers as preferred split points. BGE-M3 embeds all chunks in a single batched call returning both dense and sparse vectors. Qdrant upserts in batches of 50, each point carrying both vector types. PostgreSQL bulk-inserts all chunk records.

4.4 LangGraph Query Pipeline — 7-Node StateGraph

Every user question enters a LangGraph StateGraph and flows through up to seven nodes before returning a structured JSON answer.

Node	Function
router_node	Claude 3.5 Sonnet classifies query: 'blueprint' | 'spec' | 'cross_document'
blueprint_retrieve	Regex detects explicit page number → direct PostgreSQL fetch (bypasses Qdrant). Otherwise: ColQwen2 embeds query → Qdrant MaxSim top-20
text_retrieve	BGE-M3 embeds query (dense + sparse) → Qdrant RRF fusion (prefetch dense top-100 + prefetch sparse top-100)
rerank_node	BGE-Reranker-v2-M3 cross-encoder on text-rich docs; ColQwen2 score fallback for visual-only pages
confidence_check	Avg top-3 reranker scores normalised [0,1]. >=0.4 → generate. <0.4 → rewrite. 0 results → widen. Max 2 retries.
rewrite_node	Claude rewrites low-confidence query to be more specific; increments retry_count
widen_node	Switches query_type to cross_document, clears retrieved_docs, retries both branches in parallel
generate_node	3x3 vision tiling → 18 base64 JPEG tiles → Claude 3.5 Sonnet → structured markdown answer with tile-level citations

4.5 The 3x3 Vision Tiling Innovation

The most significant technical contribution of Week 3. Each retrieved blueprint page is divided into a 3x3 grid of 9 tiles rather than sent as a single full-page image. Each tile covers 1/3 of the

page width and height, is upscaled to 2048px wide using LANCZOS resampling, and JPEG-encoded at quality 92 before base64 encoding.

The top 2 retrieved pages are tiled, producing 18 images per LLM call. Each tile is sent as an interleaved text label + base64 image_url block, labeled: top-left, top-center, top-right, middle-left, middle-center, middle-right, bottom-left, bottom-center, bottom-right.

Metric	2x2 Grid (Before)	3x3 Grid (After)
Tiles per page	4	9
Images per LLM call	12 (3 pages x 4)	18 (2 pages x 9)
Avg query latency	~35s	~25–26s (faster)
Schedule door table	8-12: W=16, H=10 (wrong)	8-12: W=12, H=7 (correct)
Angle restriction formulas	Generic description	Exact: L<X: 15deg, X<L<2X: 30deg, L>3X: no restriction
Flex duct max length	Not found	MAX. 8'-0 LENGTH FULLY EXTENDED
Hanger strap specs	Not found	MIN. 1 WIDE HANGER STRAP, MIN. 1.5 BAND CLAMP
Condensing unit clearance	Generic	6 ABOVE GRADE, 6 ABOVE FINISHED FLOOR
Confidence score (page 24)	~0.95	1.0

Why it is faster despite more tiles: the 3x3 system uses 2 source pages instead of 3, and smaller per-tile pixel count means lower token cost per LLM call. The net result is both better accuracy and lower latency.

4.6 DPI Upgrade: 150 DPI to 250 DPI

Early in Week 3 a critical quality issue was identified. At 150 DPI, annotation text on blueprint pages was 1-2 pixels tall — completely unreadable by the vision model. Upgrading to 250 DPI increased file sizes from ~1MB to ~6.3MB per page but made annotation text 3-5 pixels tall, readable by Claude. The vision input max resolution was also raised from 1024px to 2048px at quality 92%.

Parameter	Before	After
DPI	150	250
File size (page_8.jpg)	~1MB	~6.3MB
Annotation text size	1-2px (unreadable)	3-5px (readable)
Vision input max resolution	1024px	2048px
JPEG quality	85%	92%

4.7 System Prompt Engineering

The generate_node system prompt was iteratively refined throughout the week. The final version instructs Claude to perform exhaustive tile-by-tile extraction regardless of query length — a short query like 'explain page 4' triggers the same full response as an explicit detailed request. Claude is told to:

- Process every tile — never skip, even if visually empty
- Transcribe every word verbatim — labels, notes, callouts, annotations
- Read every dimension string exactly — fractions like 3'-6", scales like 1/8"=1'-0"
- Describe every diagram and trace every leader line to what it points at
- Read every row and column of every table or schedule
- Always include the full title block: drawing number, project address, issue date, firm name, drawn-by, checked-by

Response format: structured JSON with answer (markdown), cited_sources (with tile labels like 'Blueprint page 24 — middle-center tile'), and confidence_score.

5. Test Results — 52-Page Blueprint Stress Test

5.1 Initial 12-Question Stress Test

A 52-page real construction blueprint (Airport Office Building Improvements, Naples FL) was uploaded and 12 questions were run spanning all query types.

Q#	Query	Conf	Latency	Result
Q01	Building footprint dimensions	0.70	60.5s	Partial — scale bar found, true L x W not dimensioned
Q02	Structural system type	0.70	49.4s	Steel frame identified correctly
Q03	Column grid spacing	0.90	36.4s	5 grid sizes extracted from page 23
Q04	HVAC type and duct sizes	0.80	47.0s	VAV confirmed, no CFM visible (honest)
Q05	Electrical panel specs	0.95	31.0s	Panel A/B/M — amperage, voltage, location all correct
Q06	Plumbing fixture schedule	0.95	38.2s	Per-room fixture counts, both floors
Q07	Exterior wall assembly	0.70	33.3s	Materials listed, R-values not present (correct)
Q08	Finish schedule / flooring	0.95	39.9s	Brand-specific: Mannington, Patcraft, Trinity Tile
Q09	Floor-to-ceiling heights	0.70	51.2s	Partial — counter heights found, slab-to-slab not dimensioned
Q10	Door schedule	0.98	36.7s	Full schedule: sizes, hardware sets, fire ratings, glazing
Q11	ADA / accessibility features	0.90	45.3s	Routed to tech sheet — limited data (honest redirection)
Q12	All dimensions > 50 ft	0.95	34.4s	Found 75ft/100ft egress callout across all pages

Average latency: **~42 seconds per query**

Average confidence: **0.86**

Strong answers (no partial): **8 / 12**

Partial answers: **4 / 12 (all correctly flagged as partial)**

5.2 Extended 15-Question Test (Both Pipelines)

After adding the specification ingestion pipeline and upgrading to 3x3 tiling, a second 15-question test was run. This test also exercised cross-document queries combining visual and text results.

Q#	Type	Query	Conf	Quality
Q01	Visual/ Geometric	Page 23 room dimensions	0.55	Boardroom 18x18, offices 12x12 to 30x24
Q02	Spec/Text	Demolition notes	0.95	Detailed written requirements from spec chunks
Q03	Cross	HVAC duct sizes	0.90	Visual page 24 tiles + spec chunks combined
Q04	Geometric	Full door schedule	0.73	Complete: mark/width/height/type/material/ glazing/frame/hardware/rating
Q05	Spec	Wall assembly materials	0.76	Metal stud, 5/8 GWB, sound batt, pocket door details
Q06	Visual/ Schedule	Electrical panel schedules	0.63	DP-A 800A, Panel A/B/M/SP2 with full circuit data
Q07	Geometric	Ceiling heights	0.58	Honest: detail dimensions found, room-by-room heights not dimensioned
Q08	Spec+Visual	Plumbing fixture counts	0.73	Per-room: WC/urinal/lav counts for floors 1 and 2
Q09	Spec	Fire protection specs	0.70	Design density, GPM, NFPA 13/25 refs, sprinkler head schedule
Q10	Page-specific (short)	What is on page 24	0.54	Full 9-tile: duct access panel, flex duct, louver, condensing unit
Q11	Cross	Structural column/beam sizes	0.66	Honest 'not in this set' — redirected to S-series drawings
Q12	Spec+Visual	Interior wall assemblies	0.49	Metal stud, 5/8 GWB, sound batt, pocket door details
Q13	Page-specific (short)	What is on page 4	0.55	Full 9-tile: life safety, egress distances, wall ratings, code refs
Q14	Cross	Demolition scope	0.71	First + second floor scope, FAA/data equipment continuity
Q15	Visual/ Schedule	HVAC equipment schedule	0.79	Full Carrier/Mitsubishi schedule: model#, BTU, SEER, CFM

Notable observations:

- Short page queries ('what is on page 4', 'what is on page 24') trigger exhaustive 9-tile extraction including full title blocks
- Cross-document queries correctly combine ColQwen2 visual results and BGE-M3 text chunk results in a single answer
- The pipeline returns honest 'not found' answers when data genuinely is not in the document set (structural drawings, slab heights)
- Electrical and mechanical equipment schedules are extracted with full table fidelity including brand names and model numbers
- Confidence scoring is calibrated: spec-only queries score 0.90+, visual-only queries score ~0.55–0.65

6. Gemini Flash vs. RAG Pipeline — Architecture Comparison

6.1 How They Work

Approach A — RAG Pipeline (What We Built)

Chunk and index the document upfront. At query time retrieve only the most relevant pages and send those to the LLM. The LLM never sees the full document — only the top 2–3 most relevant pages as tiled images.

PDF → chunk/embed/index → stored in Qdrant → user query → retrieve relevant chunks → LLM answers from those chunks only

Approach B — Gemini Flash 'Read Everything'

Skip indexing entirely. At query time convert the entire PDF to images and dump every page directly into the LLM's context window (up to 1M tokens). No retrieval layer.

PDF → convert all pages to images → user query → send ALL pages + question to LLM → LLM answers directly

6.2 Detailed Comparison

Dimension	Gemini Flash (Full Document)	RAG Pipeline (Ours)
Query latency	Slow — 200 images per query	Fast — 18 tiles per query (~25-42s)
Accuracy on blueprints	High — sees full document, cross-page reasoning	High — depends on retrieval quality
Cost per 100 questions	~$20 (200 pages x $0.001/page x 100)	~$1.25 (5,000 tokens avg x 100 queries / 1M)
Document size limit	~1M tokens (burns fast with images)	No limit — index once, query forever
Setup complexity	Near zero — convert PDF, call API	Weeks of engineering (what we built)
Freshness	Always reads current file	Requires re-indexing on document change
Spatial blueprint understanding	Excellent — full sheet visible at once	Good — ColQwen2 patch vectors preserve spatial info
Multi-doc Q&A	Impractical — too many tokens	Native — filter by document_id
Production scalability	Cannot scale — cost explodes	Scales well — fixed cost per query

6.3 Cost Math

Gemini Flash (full document, 200 pages per document):

- Cost per query: 200 pages x $0.001/page = $0.20 per question
- 100 questions: $20.00
- 1,000 questions: $200.00 — not viable for production

RAG Pipeline (top-2 pages, ~5,000 tokens per query):

- Cost per query: ~$0.0125 per question (Claude Sonnet pricing, ~5K tokens)
- 100 questions: ~$1.25
- 1,000 questions: ~$12.50 — 16x cheaper than Gemini Flash at scale

6.4 Recommended Hybrid Architecture

The optimal production strategy is a hybrid that uses the right tool for each task:

Task	Recommended Approach	Reason
Full-document report generation	Gemini Flash — send all pages	One-time task, no Q&A loop, full context needed
Targeted Q&A in production	RAG Pipeline (what we built)	Cheap, fast, scalable, handles multi-doc
Page retrieval	ColQwen2 RAG to find relevant pages	Then send those pages as images to LLM
Spec document search	BGE-M3 hybrid RAG	Text chunking works well for written specs

In practice: use Gemini Flash to generate a one-time intake report when a contractor uploads a new document. Then use the RAG pipeline for all subsequent Q&A against that document. This saves money — the expensive Gemini Flash call happens once per document, not once per question.

7. Future Enhancements & Roadmap

7.1 RAGAS Evaluation Framework (Next Priority)

RAGAS (Retrieval Augmented Generation Assessment) has already been added to the dependency list (ragas 0.1.21). The next immediate step is wiring it up to measure the pipeline's performance rigorously rather than relying on manual inspection.

RAGAS will evaluate:

- Faithfulness — does the answer contain only claims supported by the retrieved context?
- Answer Relevancy — does the answer address the question?
- Context Precision — are the retrieved pages actually relevant to the question?
- Context Recall — did retrieval find everything needed to answer the question?

The test suite added this week (test_query_intents.py) provides the question-answer pairs needed to run RAGAS offline. Plan: run RAGAS against the 52-page blueprint with the current pipeline to get a baseline score, then use it to measure improvement as each enhancement is applied.

7.2 HuggingFace Inference Endpoints — Faster Local Models

The biggest current bottleneck is model inference speed. ColQwen2 and BGE-M3 running on CPU take 25–60 seconds per query and 10+ minutes to ingest a 52-page document. Two paths forward:

7.2.1 HuggingFace Inference API (Cloud)

HuggingFace provides a hosted inference API for models like BGE-M3. Instead of running BGE-M3 locally, embed API calls to HF's servers. This offloads the CPU bottleneck entirely at the cost of API latency and token cost. For BGE-M3 specifically, HF's inference endpoint is well-optimized and has very low latency on their hardware.

The HF_API_KEY environment variable is already in the pipeline — the switch is a matter of swapping the embedding call from local BGEM3FlagModel to a HuggingFaceInferenceClient call.

7.2.2 Quantization (Local)

For running faster on the development machine:

- BGE-M3: INT8 quantization reduces model size by ~4x with minimal accuracy loss on retrieval tasks. FlagEmbedding supports quantized inference.
- BGE-Reranker: INT8 quantization similarly applicable.

- ColQwen2: More constrained — quantization interacts with the multi-vector architecture. float16 would be ideal but MPS bfloat16 is broken on Apple Silicon. This requires testing on the specific matmul shapes used.

7.2.3 Enabling Apple Silicon MPS (Research Item)

ColQwen2 on MPS fails due to a bfloat16 matmul shape incompatibility. A potential fix is to cast the model to float16 (rather than bfloat16) before loading to MPS, or to patch the specific matrix multiplication that fails. If MPS can be enabled for ColQwen2, ingestion time on Apple Silicon would drop from 10+ minutes to under 2 minutes for a 52-page document.

7.3 Intent-Based Query Routing (Partially Built)

An enhanced intent classifier was designed and partially implemented this week. Instead of three broad categories (blueprint / spec / cross_document), the router detects five specific intents:

- PAGE_SPECIFIC — query contains page number or sheet reference; bypasses Qdrant and fetches directly from PostgreSQL
- TABLE_SCHEDULE — query about schedules, panels, equipment lists; generates with schedule-separation enforcement
- CROSS_PAGE_FACT — general multi-page factual query
- ABSENCE_CHECK — query about content likely not in the document set; caps confidence at 0.35
- GEOMETRY_VISUAL — query about spatial layout, dimensions, locations

This is the next architectural upgrade after RAGAS baseline measurement.

7.4 Confidence Calibration Fix

The current confidence scoring has a known issue: it returns 0.95 or 1.0 even when attribution is weak or cited_sources are empty. This is because the confidence is derived purely from reranker scores, not from answer quality. A revised confidence function has been designed that applies penalties for: fewer than 2 cited sources (-0.20), PAGE_SPECIFIC query where the page was not found directly (-0.15), ABSENCE_CHECK intent (cap at 0.35), and very short answers (-0.15). The hard cap is 0.90 — a score of 1.0 will never be returned.

7.5 GraphRAG Exploration (Longer Horizon)

GraphRAG (knowledge graph-augmented retrieval) is on the longer-horizon research list. For construction documents, a knowledge graph could encode relationships like 'Panel A is on floor 2', 'AHU-2-5 serves the Conference Room', 'fire protection zone FP201A covers both floors' — enabling multi-hop reasoning queries like 'what mechanical systems are in rooms served by Panel A?' that current flat vector search cannot answer well.

This is a research exploration item for a later phase, not the immediate roadmap.

7.6 Output Schema Improvement

The answer JSON schema is being extended to include:

- answer_type — page_summary | structured_schedule | cross_page_summary | abstention | partial
- confidence_rationale — one sentence explaining the confidence score
- cited_sources — enhanced with page_number (int), sheet_title, and evidence_type (text | visual | schedule)
- is_partial — boolean flag
- missing_information — what could not be found, or null

The Chainlit UI will surface answer_type as a label, confidence_rationale as a subtitle, and missing_information as a note at the bottom of each response.

8. Known Constraints & Open Issues

Issue	Status	Planned Fix
ColQwen2 CPU-only (MPS bfloat16 broken)	Active constraint	Research float16 MPS patch; or use HF Inference API
Ingestion latency 10+ min for 52 pages	Active constraint	Quantization or HF Inference API
Query latency 25–60s	Active constraint	HF Inference API for embeddings + reranker
Confidence returns 0.95–1.0 even on weak answers	Known bug	Confidence calibration redesign (Section 7.4)
Schedule merging across panels on same page	Known bug	Schedule boundary enforcement in system prompt (planned)
cited_sources empty in some responses	Known bug	Output schema fix + confidence penalty
torch + torchvision must match exactly (2.6.0 / 0.21.0)	Gotcha	Document in setup guide; version-pin in requirements.txt
Chainlit must be >=2.0.0 (1.x breaks with FastAPI)	Gotcha	Already pinned
No auth on API endpoints	By design (local/dev)	Add API key middleware before production
RAGAS evaluation not yet wired up	Next step	Wire up ragas 0.1.21 against existing test suite

9. Summary — What Was Built This Week

In three weeks the PlanSmartAI pipeline went from a framework selection decision to a working multimodal RAG system tested against a real 52-page construction blueprint. Here is what was built:

Week 1

- Selected LangGraph + LangChain as the agentic orchestration framework
- Selected Qdrant as the vector database (only DB supporting both MultiVector and Sparse for the two-collection architecture)
- Defined the two-pipeline architecture (visual for blueprints, text for specs)

Week 2

- Built and validated baseline text retrieval pipeline
- Researched and selected ColQwen2 for visual embedding (patch-level spatial preservation)
- Researched and selected BGE-M3 for dual dense+sparse text embedding
- Researched and selected BGE-Reranker-v2-M3 for cross-encoder reranking
- Identified Apple Silicon MPS constraint and confirmed CPU fallback path

Week 3

- Built full LangGraph StateGraph with 7 nodes: router, 2 retrieval branches, rerank, confidence_check, rewrite, widen, generate
- Implemented 3x3 vision tiling — 9 tiles per page at 2048px, 18 images per LLM call
- Upgraded DPI from 150 to 250 — annotation text now readable
- Engineered exhaustive tile-extraction system prompt
- Ran 27 test queries across 2 test suites — 8/12 strong answers, 4/12 honest partials in first test; all 15 passed quality bar in second test
- Designed intent-based router upgrade, confidence calibration fix, and output schema enhancement
- Documented full cost comparison: RAG pipeline is 16x cheaper than Gemini Flash at scale
- Added RAGAS dependency — evaluation wiring is the next step